

docker-compose

Usuario

http://

**frontend**
:5173


React + Vite + TypeScript





**FastAPI**
:8000

/api/osrm

/api/otp

/api/gtfs

/api/lpmc

**osrm-backend**
:latest

**osrm-car**
:5000

**osrm-bike**
:5001

**osrm-foot**
:5002

**otp**
:8080

**OpenTripPlanner**
Multimodal Trip Planning



/predict

**XGBoost**

/train

**Random Forest**

**Keras**

Fuentes de datos

**GEOFABRIK**

**OpenStreetMap**
Castilla-La Mancha
CLM.osm.pbf

osrm-extract (x3)

**Grafos OSRM**
car / bike / foot

OTP build

**graph.obj**

Feed GTFS
stops.txt · routes.txt · trips.txt

Descargar GTFS

**Bus Urbano Toledo**

**NAP**
Transporte Multimodal

**Dataset LPMC**
Hillel et al. 2018 – Londres

**Transport for London**

**EPFL**

